**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING • PES UNIVERSITY, BENGALURU**

*UE24CS395: Database Management Systems Mini Project • Academic Year 2025–2026*

# Relational Schema with Keys, Relationships & Constraints

**Evaluation Matrix Row 2: Relational Schema • Weightage: 3 Marks**

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## 1. ER-to-Relational Mapping Methodology

We transformed our conceptual ER model into a relational database schema by applying the standard 4-step mapping algorithm:

1. **Step 1: Mapping Strong Entities:** For each strong entity (`officers`, `cases`, `people`, `organizations`, `vehicles`, `devices`, `locations`, `evidence`), we created a relation containing all atomic attributes, with the designated primary key enforced via `PRIMARY KEY` constraints.
2. **Step 2: Mapping Weak Entities:** For weak entities (`evidence_custody`, `case_activity`), we created relations including all local discriminator attributes plus the primary key of the owner parent relation as a foreign key with `NOT NULL` constraints.
3. **Step 3: Mapping 1:N Binary Relationships:** For 1:N relationships (e.g. Lead Officer on Cases, Case on Evidence), the primary key of the "1-side" entity is placed as a foreign key column in the "N-side" relation (e.g. `cases.lead_officer_id` referencing `officers.officer_id`).
4. **Step 4: Mapping M:N Binary Relationships:** For each of the 12 many-to-many relationships, we created a distinct associative relation whose composite primary key is formed by the primary keys of the participating entities, augmented by relationship-specific descriptive columns.

## 2. Complete 22-Table Relational Schema Catalog

Table 1 catalogs all 22 tables in the active schema, documenting primary keys, foreign keys, unique and check constraints, and cardinalities.

| Table Name | Primary Key | Foreign Keys (References) | Constraints (CHECK / UNIQUE) | Cardinality & Purpose |
|---|---|---|---|---|
| officers | officer_id | None | badge_number UNIQUE, email UNIQUE | 1:N with cases, evidence, custody, activity. |
| cases | case_id | lead_officer_id → officers | CHECK(status IN (...)), CHECK(priority IN (...)) | Core investigative dossier; 1:N with exhibits/logs. |
| people | person_id | None | CHECK(category IN (...)) | Implicated persons, suspects, witnesses. |
| organizations | organization_id | None | registration_number UNIQUE | Corporate entities, commercial syndicates. |
| vehicles | vehicle_id | None | registration_number UNIQUE | Automotive assets implicated in cases. |
| devices | device_id | None | serial_number UNIQUE | Hardware seized for digital forensics. |
| locations | location_id | None | CHECK(lat BETWEEN -90 AND 90) | Geospatial incident & storage locations. |
| evidence | evidence_id | case_id → cases, collected_by → officers | barcode UNIQUE, CHECK(status IN (...)) | Seized forensic exhibits; 1:N with custody log. |
| evidence_custody | custody_event_id | evidence_id → evidence, officer_id → officers | verification_hash NOT NULL | Weak audit relation; immutable custody chain. |
| case_activity | activity_id | case_id → cases, officer_id → officers | activity_timestamp NOT NULL | Weak audit relation; automated trigger log. |
| case_people | (case_id, person_id) | case_id → cases, person_id → people | Composite PK | M:N associative junction (Case ↔ Person). |
| case_organizations | (case_id, org_id) | case_id → cases, org_id → organizations | Composite PK | M:N associative junction (Case ↔ Organization). |
| case_vehicles | (case_id, vehicle_id) | case_id → cases, vehicle_id → vehicles | Composite PK | M:N associative junction (Case ↔ Vehicle). |
| case_devices | (case_id, device_id) | case_id → cases, device_id → devices | Composite PK | M:N associative junction (Case ↔ Device). |
| case_locations | (case_id, loc_id) | case_id → cases, loc_id → locations | Composite PK | M:N associative junction (Case ↔ Location). |
| person_organizations | (person_id, org_id) | person_id → people, org_id → organizations | Composite PK | M:N associative junction (Person ↔ Org membership). |
| person_vehicles | (person_id, vehicle_id) | person_id → people, vehicle_id → vehicles | Composite PK | M:N associative junction (Person ↔ Vehicle ownership). |

| Table Name | Primary Key | Foreign Keys (References) | Constraints (CHECK / UNIQUE) | Cardinality & Purpose |
|---|---|---|---|---|
| person_devices | (person_id, device_id) | person_id → people, device_id → devices | Composite PK | M:N associative junction (Person ↔ Device usage). |
| evidence_people | (evidence_id, person_id) | evidence_id → evidence, person_id → people | Composite PK | M:N associative junction (Evidence ↔ Person link). |
| evidence_vehicles | (evidence_id, vehicle_id) | evidence_id → evidence, vehicle_id → vehicles | Composite PK | M:N associative junction (Evidence seized in vehicle). |
| evidence_devices | (evidence_id, device_id) | evidence_id → evidence, device_id → devices | Composite PK | M:N associative junction (Forensic extraction on device). |
| evidence_locations | (evidence_id, loc_id) | evidence_id → evidence, loc_id → locations | Composite PK | M:N associative junction (Seizure scene location). |

## 3. Normalization Analysis & Formal Proofs (1NF to BCNF)

We systematically evaluated the functional dependencies (FDs) of our relations to ensure strict adherence to Boyce-Codd Normal Form (BCNF).

### 3.1 First Normal Form (1NF) Compliance

A relation is in 1NF if and only if all domain values are atomic and there are no repeating groups.

- **Violation Avoidance:** Rather than storing aliases or phone numbers as multivalued arrays or comma-delimited strings within `people`, distinct phone numbers and address lines are maintained in dedicated columns, and person-to-asset linkages are isolated in 1NF associative relations.
- **Atomicity:** Coordinates in `locations` are stored as separate atomic `NUMERIC(10,8)` latitude and `NUMERIC(11,8)` longitude values, not combined geometric strings.

### 3.2 Second Normal Form (2NF) Compliance

A relation is in 2NF if it is in 1NF and every non-prime attribute is fully functionally dependent on the entire primary key (no partial dependencies).

- **Single-Attribute Keys:** All 10 entity relations (e.g. `cases`, `officers`, `people`) possess single-attribute primary keys (e.g. `case_id`). Therefore, partial functional dependencies are mathematically impossible.
- **Composite Key Junctions:** In our 12 associative relations (e.g. `case_people` with composite PK `{case_id, person_id}`), the non-key attributes are `role_in_case`, `notes`, and `added_at`. These attributes describe the specific relationship between the case and the person:
  `{case_id, person_id} → {role_in_case, notes, added_at}`
  Neither `case_id → role_in_case` nor `person_id → role_in_case` holds individually because a person may be a "Witness" in one case and a "Suspect" in another. Thus, full functional dependency is preserved.

### 3.3 Third Normal Form (3NF) & BCNF Compliance

A relation is in 3NF if it is in 2NF and no non-prime attribute is transitively dependent on the primary key. A relation is in BCNF if for every non-trivial functional dependency `X → Y`, `X` is a superkey.

Consider relation `evidence`:

**Functional Dependencies for Relation EVIDENCE**

```
FD1: evidence_id → {case_id, title, evidence_type, description, status, storage_location,
                    chain_of_custody_hash, barcode, collected_by_officer_id, created_at,
updated_at}
FD2: barcode → {evidence_id, ...}  (Candidate Key via UNIQUE constraint)
```

In `evidence`, the only determinants are `evidence_id` and `barcode`. Both are candidate keys. We do not store officer names or case titles in `evidence` (which would introduce transitive dependencies `evidence_id → case_id → case_title`). Instead, such details are resolved dynamically via relational JOINs.

Therefore, every determinant in the schema is a superkey, and the schema satisfies **Boyce-Codd Normal Form (BCNF)**.